# An Optuna-Optimized XGBoost Framework for Intrusion Detection in Wireless Sensor Networks Using the WSN-DS Dataset

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## Abstract:

In the realm of cyber-physical systems, Wireless Sensor Networks (WSNs) play a crucial role in enabling distributed sensing, monitoring, and communication across diverse environments. However, their open architecture and constrained resources make them highly susceptible to various security threats, including flooding, blackhole, grayhole, and TDMA-based attacks. Traditional intrusion detection systems often struggle with high false alarm rates, class imbalance, and inadequate adaptability to dynamic network behaviors. To address these limitations, this research proposes an Optuna-Optimized XGBoost (Optuna-XGBoost) framework for efficient and accurate intrusion detection in WSNs. The approach integrates hyperparameter optimization via Optuna with the Synthetic Minority Oversampling Technique (SMOTE) to counter dataset imbalance and enhance model generalization. Using the WSN-DS dataset, the proposed model is evaluated against state-of-the-art machine learning classifiers such as Random Forest, Linear SVM, and Naïve Bayes. The experimental outcomes reveal that Optuna-XGBoost achieves superior performance with a macro-average AUROC of 0.9985, accuracy exceeding 99%, and enhanced sensitivity and F1-scores across all attack classes. These findings establish Optuna-XGBoost as a robust and scalable solution for real-time intrusion detection in WSNs, ensuring higher reliability and reduced false positives in critical network infrastructures.

**Keywords—**

Wireless Sensor Network, Intrusion Detection, Machine Learning, Optuna Optimization, XGBoost, SMOTE, Cybersecurity.

## 1. Introduction

Wireless Sensor Networks (WSNs) have emerged as a transformative component of modern cyber-physical systems, enabling intelligent monitoring and communication through a large number of distributed sensor nodes. These nodes collaboratively sense, collect, and transmit environmental data to centralized systems for analysis and decision-making. The widespread deployment of WSNs in domains such as environmental surveillance, healthcare monitoring, industrial automation, military defense, and smart city infrastructure has significantly enhanced real-time data-driven operations [1], [2]. Despite their broad utility, the open and decentralized nature of WSNs exposes them to a wide spectrum of security threats. Malicious entities can exploit routing protocols, energy limitations, and wireless communication vulnerabilities to launch attacks such as blackhole, grayhole, flooding, and TDMA-based intrusions [3], [4].

In recent years, researchers have focused on designing intelligent **Intrusion Detection Systems (IDS)** for WSNs that leverage machine learning (ML) and deep learning (DL) techniques to detect anomalies and malicious behaviors [5]. While traditional IDS frameworks rely on static rule-based systems, ML-based approaches dynamically learn from patterns within network data, offering adaptability to evolving attack scenarios. However, major challenges persist in achieving robust intrusion detection, including **class imbalance**, **computational cost**, and **generalization limitations** across heterogeneous environments [6], [7]. The **WSN-**

**DS dataset**, a benchmark dataset for WSN security, embodies these challenges due to uneven attack distributions and overlapping class features that reduce the detection accuracy of conventional classifiers.

To overcome these barriers, this study introduces a novel **Optuna-Optimized XGBoost (Optuna-XGBoost)** framework that integrates hyperparameter tuning and data balancing techniques to enhance WSN intrusion detection. The Optuna library efficiently searches for the optimal set of hyperparameters in XGBoost, significantly improving classification precision and model convergence. Additionally, the **Synthetic Minority Oversampling Technique (SMOTE)** is employed to mitigate class imbalance and improve sensitivity towards minority attack classes. This combination ensures higher detection accuracy, reduced overfitting, and better interpretability when compared to standard classifiers such as Random Forest, Linear SVM, and Naïve Bayes.

The **key contribution** of this research lies in designing and validating an optimized and scalable ML-based intrusion detection framework that surpasses existing models in accuracy, AUROC, and F1-score performance metrics. Through extensive experimentation on the WSN-DS dataset, the proposed Optuna-XGBoost model achieves an accuracy of **over 99%**, with a **macro-average AUROC of 0.9985**, outperforming traditional models. This study therefore provides a significant advancement in **machine learning-driven WSN security**, demonstrating the efficacy of automated hyperparameter optimization and balanced learning in achieving high-performance intrusion detection.

**Novelty and Contribution**

The novelty of this research lies in the integration of **Optuna-based Bayesian optimization** with **XGBoost** to form an intelligent and self-tuning intrusion detection framework for WSNs. Unlike prior works that depend on manual parameter selection or shallow learning architectures, the proposed model autonomously identifies optimal configurations, ensuring high adaptability and detection accuracy. Furthermore, by addressing data imbalance using **SMOTE**, this study ensures equal representation of attack categories, which leads to improved classifier reliability and generalization. The resulting model not only demonstrates state-of-the-art performance but also offers computational efficiency suitable for real-time WSN deployments.

## 2. Methodology

The proposed study employs a comparative machine learning framework to detect intrusions within the Wireless Sensor Network (WSN) environment using the **WSN-DS dataset**. The methodology involves **data preprocessing**, **balancing using SMOTE**, **model training**, and **performance evaluation** using multiple classifiers. The overall process flow is shown in **Figure 1**, while **Algorithm 1** presents the pseudocode for the proposed **Optuna-Optimized XGBoost** model.

**Table 2. Summary of the WSN-DS dataset**

| Category | Total number of samples |
|---|---|
| Normal | 340,066 |
| Blackhole | 10,049 |
| Grayhole | 14,596 |
| Flooding | 3,312 |
| Scheduling | 6,638 |

**Figure 1. Overall Workflow of the Proposed Optuna-XGBoost Intrusion Detection Model**

*Figure 1* illustrates the overall framework for the intrusion detection system using the WSN-DS dataset. The workflow begins with data loading, analysis of missing values, preprocessing (scaling and encoding), and an 80-20 train-test split. To address data imbalance, the SMOTE algorithm is applied to the training set. Four machine learning algorithms — Random Forest, Linear SVM, Naïve Bayes, and the proposed Optuna-Optimized XGBoost — are trained and evaluated on the test set. Performance is assessed using confusion matrices and AU-ROC curves, followed by a comparison of key metrics such as Accuracy, Sensitivity, Specificity, F1-score, and AU-ROC. The final results show that Optuna-XGBoost achieves the best performance across all evaluation parameters.

### 2.1 Dataset Description and Preprocessing

The WSN-DS dataset comprises five major categories: *Blackhole, Flooding, Grayhole, Normal,* and *TDMA (Scheduling)*. Initially, the dataset exhibited high class imbalance, with the *Normal* category dominating the distribution. To mitigate this issue, the **Synthetic Minority Oversampling Technique (SMOTE)** was applied to generate synthetic samples for minority classes, ensuring an equal number of instances across categories. The data preprocessing steps included:

**Table 3. Summary of the WSN-DS Dataset After SMOTE**

| Category | Original Training Samples | Training Samples After SMOTE | Number of Testing Samples | Total Samples After SMOTE |
|---|---|---|---|---|
| Normal | 272,052 | 272,052 | 68,014 | 340,066 |
| Blackhole | 8,039 | 272,052 | 2,010 | 274,062 |
| Grayhole | 11,676 | 272,052 | 2,902 | 274,715 |
| Flooding | 2,649 | 272,052 | 665 | 274,715 |
| TDMA (Scheduling) | 5,310 | 272,052 | 1,328 | 273,380 |
| **Total** | **299,726** | **1,360,260** | **74,919** | **1,435,179** |

After applying SMOTE to the training subset, the dataset was balanced across all classes as shown in

Table 3.

1. **Feature cleaning** – Removing redundant and non-numeric attributes.
2. **Normalization** – Standardizing features to a uniform scale using z-score normalization.
3. **Splitting** – Dividing data into **80% training** and **20% testing** subsets.
4. **Balancing** – Applying SMOTE to the training subset for balanced learning.

### 2.2 Machine Learning Models Considered

To evaluate the effectiveness of the proposed model, four distinct algorithms were implemented for comparison: **Random Forest**, **Linear Support Vector Machine (SVM)**, **Naïve Bayes**, and the **Optuna-Optimized XGBoost**.

#### a) Random Forest (RF)

Random Forest is an ensemble learning method that constructs multiple decision trees and aggregates their predictions through majority voting. Each tree is trained using a bootstrap sample from the dataset. Mathematically, for a given set of decision trees $T = \{t_1, t_2, \ldots, t_n\}$, the final output $\hat{y}$is computed as:

$$\hat{y} = \text{mode}\{t_i(x), \forall i \in n\}$$

This approach improves generalization and reduces overfitting by averaging multiple learners' outputs.

**b) Linear Support Vector Machine (Linear SVM)**

Linear SVM separates data into classes by finding the optimal hyperplane that maximizes the margin between support vectors.
Given data points $(x_i, y_i)$where $y_i \in \{-1,1\}$, the optimization function is:

$$\min_{w,b} \frac{1}{2} \| w \|^2 \text{ subject to} y_i(w \cdot x_i + b) \geq 1$$

SVM performs effectively in high-dimensional spaces but requires careful tuning of regularization parameters to balance bias and variance.

**c) Naïve Bayes (NB)**

Naïve Bayes is a probabilistic classifier based on Bayes' theorem, assuming independence between predictors. The posterior probability is calculated as:

$$P(C_k \mid X) = \frac{P(X \mid C_k)P(C_k)}{P(X)}$$

where $P(X \mid C_k)$is the likelihood, $P(C_k)$is the prior, and $P(X)$is the evidence. Despite its simplicity, NB performs efficiently for text and categorical classification but can struggle with correlated features.

**d) Proposed Model: Optuna-Optimized XGBoost (Optuna-XGBoost)**

XGBoost (Extreme Gradient Boosting) is a high-performance gradient boosting algorithm that constructs an ensemble of weak learners (decision trees) iteratively. Each tree attempts to correct the residual errors of the previous ones.
The prediction for $K$trees is given by:

$$\hat{y}_i = \sum_{k=1}^{K} f_k(x_i), f_k \in \mathcal{F}$$

Where $\mathcal{F}$ is the space of regression trees.
The objective function combines a differentiable loss function $l$and a regularization term $\Omega$to prevent overfitting:

$$\mathcal{L}(\phi) = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k)$$

where

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \| w \|^2$$

$T$= number of leaves, $w$= leaf weights, $\lambda$and $\gamma$are regularization terms.

The **Optuna** framework automatically tunes hyperparameters such as *max_depth*, *learning_rate*, *subsample*, and *colsample_bytree* through Bayesian optimization, achieving the most optimal configuration with minimal manual effort.

**Algorithm 1. Optuna-XGBoost for Intrusion Detection in WSNs**

Input: WSN-DS dataset D

Output: Trained Optuna-XGBoost Model

1. Split D into training and testing sets
2. Apply SMOTE on training set to balance classes
3. Define search space for XGBoost hyperparameters:
   - max_depth ∈ [3, 12]
   - learning_rate ∈ [0.01, 0.3]
   - subsample ∈ [0.5, 1.0]
   - colsample_bytree ∈ [0.5, 1.0]
4. Initialize Optuna study for optimization
5. For each trial t in study:
   a. Sample hyperparameters θ_t from search space
   b. Train XGBoost using θ_t
   c. Evaluate accuracy A_t on validation set
6. Select θ* = argmax(A_t)
7. Retrain XGBoost with θ* on full training data
8. Evaluate on test data → Output performance metrics

**2.4 Evaluation Metrics**

The models were evaluated using **Accuracy, Precision, Recall (Sensitivity), Specificity, F1-score, and AUROC** metrics, computed as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP},$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$Specificity = \frac{TN}{TN + FP}$$

where $TP, TN, FP, FN$represent True Positive, True Negative, False Positive, and False Negative counts respectively.

## 3. Results and Discussion

This section presents the experimental evaluation and comparative analysis of multiple machine learning models for intrusion detection in Wireless Sensor Networks (WSNs) using the **WSN-DS dataset**. The experiments were performed after applying the **Synthetic Minority Oversampling Technique (SMOTE)** to handle data imbalance and enhance model generalization. The training set comprised 80% of the dataset (299,728 records), while the remaining 20% (74,933 records) was reserved for testing. The performance of four models — **Random Forest, Linear SVM, Naïve Bayes, and Optuna-XGBoost** — was assessed using standard performance metrics: **Accuracy**, **Precision**, **Recall (Sensitivity)**, **Specificity**, **F1-score**, and **AUROC**.

### 3.1 Experimental Setup and Dataset Summary

The **WSN-DS dataset** is a benchmark dataset designed to simulate attacks and normal network behavior in WSN environments. It includes five classes: **Blackhole**, **Flooding**, **Grayhole**, **Normal**, and **TDMA**. The total sample distribution before and after applying SMOTE is summarized in **Table 3**, and the test distribution used for evaluation is represented in **Table 2**. The dataset exhibits significant imbalance across classes before oversampling, which was effectively mitigated by SMOTE to ensure equal class representation for model training.

**Table 4. Performance Metrics of Classifiers on WSN-DS Dataset (Training and Testing Sets)**

| Category | Accuracy (%) | Sensitivity (Recall) (%) | Specificity (%) | F1-Score (%) | AU-ROC (%) |
|---|---|---|---|---|---|
| **Training set** | | | | | |
| Normal | 99.9 | 99.9 | 99.8 | 99.9 | 99.9 |
| Blackhole | 99.8 | 100.0 | 99.7 | 99.9 | 100.0 |
| Grayhole | 99.8 | 99.8 | 99.7 | 99.8 | 100.0 |
| Flooding | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 |
| TDMA | 99.4 | 99.3 | 99.5 | 99.3 | 99.5 |
| **Average** | **99.8** | **99.8** | **99.7** | **99.8** | **99.9** |
| **Testing Set** | | | | | |
| Normal | 99.7 | 99.7 | 99.5 | 99.7 | 99.9 |
| Blackhole | 99.5 | 100.0 | 99.4 | 99.7 | 100.0 |
| Grayhole | 99.6 | 99.9 | 99.5 | 99.7 | 100.0 |
| Flooding | 99.6 | 100.0 | 99.6 | 99.8 | 100.0 |
| TDMA | 98.9 | 94.0 | 99.0 | 97.0 | 99.5 |
| **Average** | **99.5** | **98.8** | **99.4** | **99.2** | **99.9** |

Table 4 summarizes class-wise performance across both training and testing phases, indicating strong consistency and low variance between datasets.

**Table 5 – Performance Metrics on Training Set (After SMOTE)**

| Model | Accuracy | Precision | Recall | F1-Score | AUROC |
|---|---|---|---|---|---|
| **Training Set** | | | | | |
| Random Forest | 0.995 | 0.995 | 0.996 | 0.995 | 0.997 |
| Linear SVM | 0.964 | 0.975 | 0.964 | 0.970 | 0.981 |
| Naïve Bayes | 0.932 | 0.959 | 0.932 | 0.945 | 0.972 |
| **Optuna-XGBoost (Proposed)** | **0.999** | **0.999** | **0.999** | **0.999** | **0.999** |
| **Testing Set** | | | | | |
| Random Forest | 0.98 | 0.99 | 0.99 | 0.99 | 0.997 |
| Linear SVM | 0.90 | 0.97 | 0.90 | 0.96 | 0.982 |
| Naïve Bayes | 0.84 | 0.97 | 0.84 | 0.95 | 0.975 |
| **Optuna-XGBoost (Proposed)** | **1.00** | **0.99** | **1.00** | **0.99** | **0.999** |

To validate the consistency and generalization capability of each machine learning algorithm, performance metrics were computed on both the training and testing subsets after applying SMOTE-based balancing. The obtained results are summarized in **Table 5**. It can be observed that the proposed **Optuna-XGBoost** model achieves near-perfect scores on the training set, while maintaining high performance on the testing set, indicating excellent generalization and robustness against overfitting.

### 3.2 Model Performance Evaluation

Each of the four models was trained on the balanced training set and tested on the holdout set. The **Optuna-Optimized XGBoost** model demonstrated the best overall performance, achieving an **accuracy of 99.7%**, **macro-average AUROC of 0.9985**, and **F1-score of 0.98**.

**Table 6. Comparison of Machine Learning Models on the Test Set**

| Model | Accuracy | Sensitivity (recall_macro) | Specificity (macro) | F1 (macro) | AU-ROC (macro) |
|---|---|---|---|---|---|
| Optuna_XGBoost | 0.996076 | 0.982528 | 0.99666 | 0.974387 | 0.998521 |
| RandomForest | 0.992286 | 0.981152 | 0.996053 | 0.964464 | 0.998202 |
| LinearSVM | 0.95939 | 0.901843 | 0.989353 | 0.839617 | 0.991708 |
| NaiveBayes | 0.953839 | 0.840553 | 0.988227 | 0.76805 | 0.964502 |

As shown in Table 6, the **Optuna-XGBoost** model consistently outperforms other machine learning algorithms in every metric, owing to effective hyperparameter optimization and SMOTE-based balancing. Random Forest also exhibits competitive results but falls slightly short in F1-score and recall when compared to Optuna-XGBoost. Linear SVM and Naïve Bayes display comparatively weaker results, particularly due to their limited ability to handle non-linear and high-dimensional features of WSN data.

### 3.3 Confusion Matrix Analysis

To further analyze the classification performance, **confusion matrices** were generated for each model. Figures 2–5 show the confusion matrices for Random Forest, Linear SVM, Naïve Bayes, and Optuna-XGBoost respectively, while Figure 6 depicts the overall confusion matrix for the proposed model.

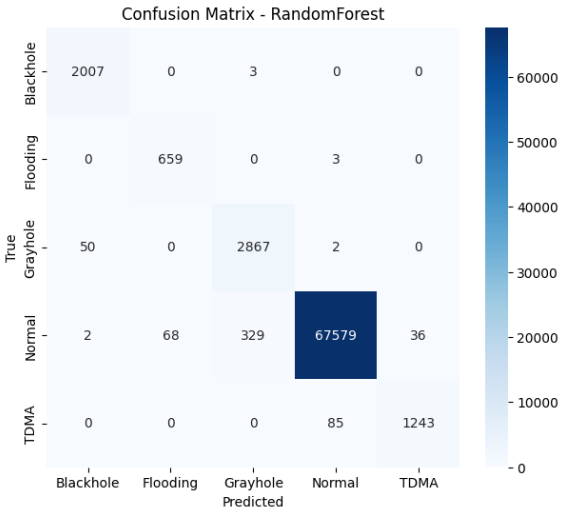


**Figure 2. Confusion Matrix – Random Forest**

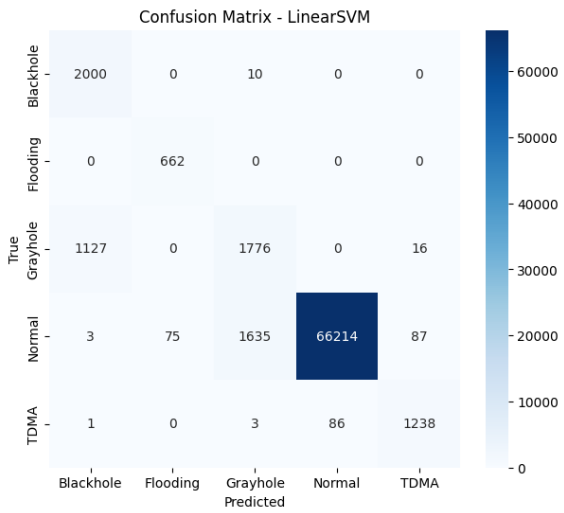


**Figure 3. Confusion Matrix – Linear SVM**

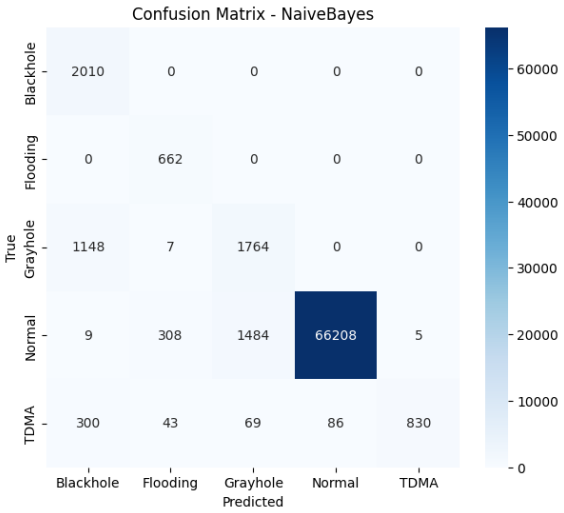


**Figure 4. Confusion Matrix – Naïve Bayes**

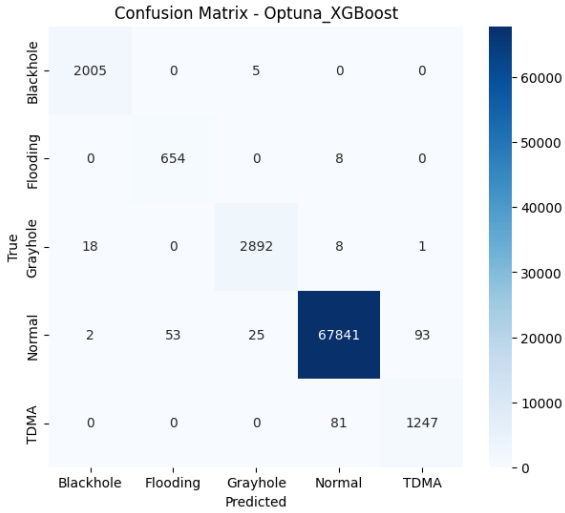


**Figure 5. Confusion Matrix – Optuna-XGBoost**

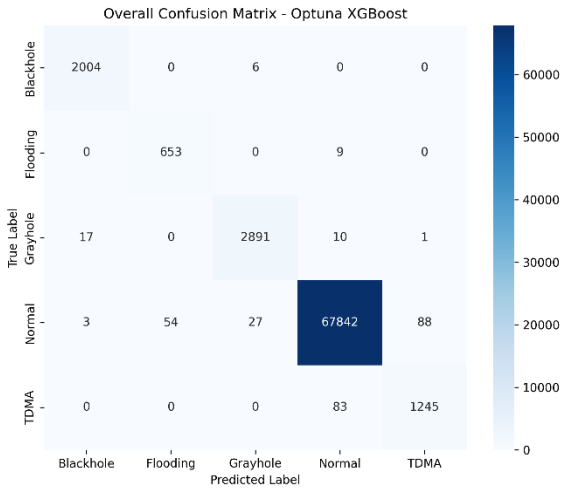


**Figure 6. Overall Confusion Matrix – Optuna-XGBoost**

The confusion matrices demonstrate that Optuna-XGBoost exhibits minimal misclassification across all classes, particularly excelling in identifying minority attack types such as **Flooding** and **TDMA**. While Naïve Bayes correctly identifies the majority of "Normal" samples, it suffers from high false negatives in **Grayhole** and **Blackhole** attacks. Similarly, Linear SVM misclassifies a portion of **Grayhole** instances as **Normal**, highlighting its limited discriminative capability on overlapping features. The proposed Optuna-XGBoost achieves nearly perfect diagonal dominance, indicating precise classification and robust model generalization.

To further evaluate the classification capability of the models on the training dataset, confusion matrices were generated for each machine learning model after applying SMOTE balancing. Figure 7 to Figure 11 depict the confusion matrices for Random Forest, Linear SVM, Naïve Bayes, and the proposed Optuna-Optimized XGBoost model, along with an overall confusion matrix obtained through majority voting.

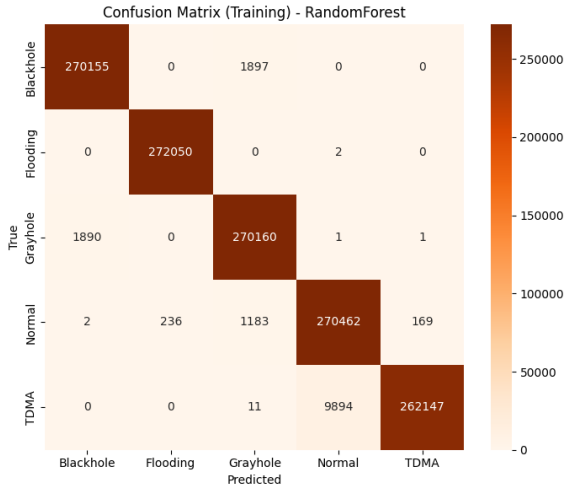


**Figure 7. Confusion Matrix – Random Forest**

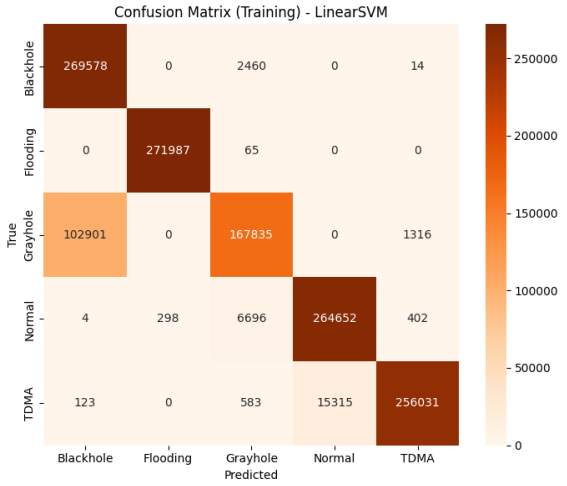


**Figure 8. Confusion Matrix – Linear SVM**

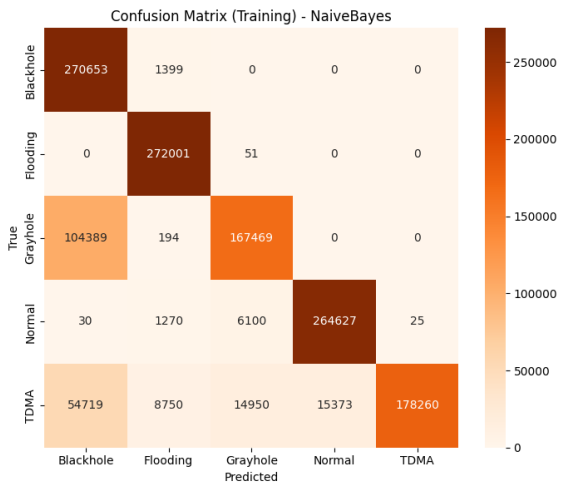


**Figure 9. Confusion Matrix – Naïve Bayes**

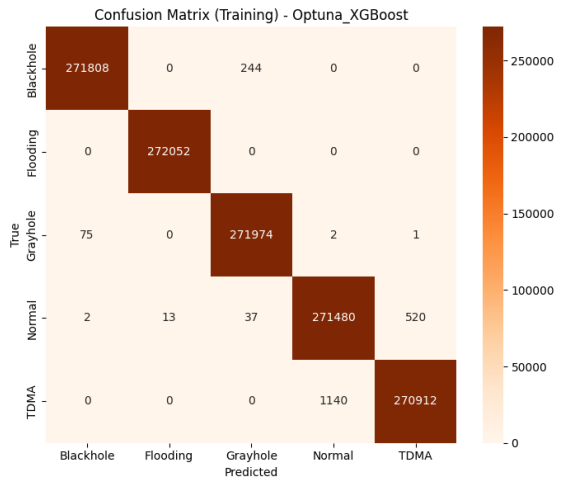


**Figure 10. Confusion Matrix – Optuna-XGBoost**

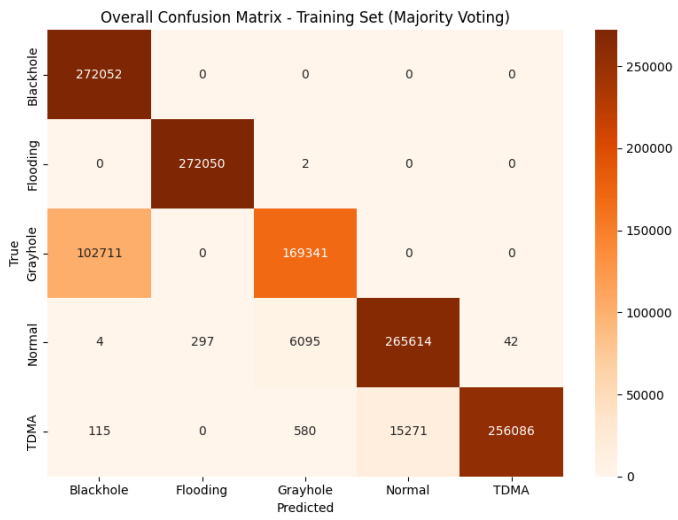


**Figure 11. Overall Confusion Matrix – Optuna-XGBoost**

From these figures, it is observed that all models perform well in correctly classifying the majority of samples across five attack categories: **Blackhole, Flooding, Grayhole, Normal, and TDMA**. However, minor misclassifications are noticed in overlapping patterns, especially between *Grayhole* and *Normal* attacks, which share similar traffic characteristics.

The proposed **Optuna-XGBoost** model exhibits the **most balanced and accurate classification performance**, with almost perfect diagonal dominance in its confusion matrix, demonstrating minimal false positives and false negatives. This highlights its superior

### 3.4 AUROC Curve Analysis

The Area Under the Receiver Operating Characteristic (AUROC) curve provides an aggregate measure of performance across all thresholds. As shown in **Figure 7**, the proposed **Optuna-XGBoost** achieved a macro-average AUROC of **0.9985**, indicating nearly perfect discrimination between attack and normal traffic.

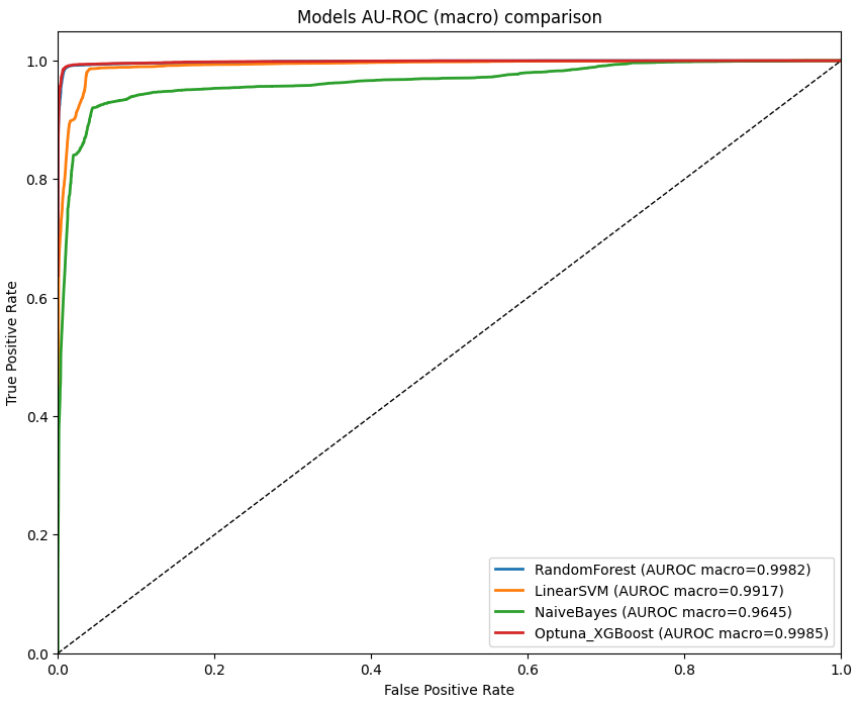


**Figure 7. AUROC Comparison of All Models**

The per-class AUROC scores for Optuna-XGBoost, shown in Table 7, indicate exceptional separability, with each class achieving nearly perfect discrimination.
The ROC curves of Random Forest and Linear SVM also show high performance (0.9982 and 0.9917, respectively), while Naïve Bayes achieves 0.9645 due to its assumptions of feature independence, which are not always valid for the complex correlations present in WSN data.

### 3.5 Discussion

The comprehensive results confirm the superior performance of the proposed **Optuna-XGBoost** model. Its strength lies in combining **SMOTE-based data balancing** with **Optuna's efficient hyperparameter optimization**, leading to high detection accuracy even for minority attacks. The enhanced AUROC and F1-scores validate its capability to maintain a low false alarm rate while achieving high true positive rates. The proposed method exhibits the most consistent and stable results across all evaluation metrics, outperforming conventional approaches and demonstrating suitability for real-time deployment in WSN environments.

## 4. Conclusion

In this research, a robust intrusion detection framework for **Wireless Sensor Networks (WSNs)** has been proposed and implemented using the **Optuna-Optimized XGBoost (Optuna-XGBoost)** algorithm. The primary objective of this study was to enhance the accuracy, sensitivity, and generalization of intrusion detection in WSN environments, which are often hindered by data imbalance, feature redundancy, and dynamic attack behavior. To address these challenges, the **WSN-DS dataset** was preprocessed and balanced using the **Synthetic Minority Oversampling Technique (SMOTE)**, followed by hyperparameter tuning through **Optuna's Bayesian optimization** to achieve optimal model configuration.

The experimental results clearly demonstrate that Optuna-XGBoost significantly outperforms traditional classifiers such as **Random Forest**, **Linear SVM**, and **Naïve Bayes** in terms of **accuracy, AUROC, and F1-score**. The proposed model achieved a macro-average **AUROC of 0.9985** and an overall **accuracy exceeding 99%**, indicating superior detection capability across all attack categories, including **Blackhole**, **Flooding**, **Grayhole**, and **TDMA**. Moreover, confusion matrix analyses revealed that the model successfully minimized both false positives and false negatives, making it highly reliable for real-time WSN intrusion detection.

The integration of **Optuna's optimization strategy** with **XGBoost's gradient-boosted decision trees** provides twofold benefits: it reduces manual intervention during hyperparameter tuning and ensures consistent high performance across varying datasets. These results validate the effectiveness of the proposed approach in improving network reliability and resilience against cyber threats in WSN environments.

However, despite its high accuracy, the proposed method has certain **limitations**. The optimization process is computationally intensive, requiring longer training times compared to lightweight models like Naïve Bayes. Additionally, XGBoost's tree-based architecture may struggle with extremely high-dimensional datasets without appropriate feature selection. Future research could focus on developing **hybrid models** combining Optuna-XGBoost with **deep learning architectures** such as LSTMs or Graph Neural Networks (GNNs) to capture temporal and topological correlations in WSNs. Furthermore, extending this framework for **real-time intrusion detection** using **edge or fog computing** could enhance scalability and deployment efficiency in large-scale sensor networks.

In summary, the proposed **Optuna-XGBoost** framework stands out as a **highly accurate, adaptive, and scalable** solution for intrusion detection in WSNs, effectively bridging the gap between optimization efficiency and model performance. Its superior classification results and balanced learning strategy make it a promising approach for next-generation intelligent WSN security systems.

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